Smart Contract Take-Home Guide

Role: Blockchain / Smart Contract Developer
Format: Paid take-home assignment
Expected duration: This is a 4-hour task

Start here

What you are looking at now:

- Branch: main
- Contents: this guide only (no source code on main)

Your coding work lives on another branch in the same repository.

Next step: choose a branch

This PDF on main is the process overview. To read the full task guide and start coding, check out one of the coding branches and open its README.md.

Repository layout (branches):

```
smart-contract/
+-- main                 <- you are here
|   +-- HOME_TASK_GUIDE.pdf
|       Process overview only (no source code)
|
+-- scratch              <- checkout to start
|   +-- README.md
|       Full requirements (build your own architecture)
|
+-- boiler-plate         <- checkout to start
    +-- README.md
    +-- contracts/
    +-- test/
    +-- Hardhat starter
        Starter code + README with what to implement
```

Important: If you still only see this PDF in the "smart-contract" folder, you are on main. Switch branches to continue.

1. Purpose

Implement a small Token Vesting Vault in Solidity with tests. Focus on correctness, safety, and clear design.

2. Process

1. Read this guide
2. Choose one option:
   - scratch - set up framework and architecture yourself

- boiler-plate - implement contract + tests in the starter

3. Build and test
4. Submit your work from this repository (or your fork)

```
git checkout scratch
# or
git checkout boiler-plate
```

3. Product summary

Create a vesting vault where:

- Admin creates a vesting schedule for a beneficiary
- Tokens unlock linearly over time
- Beneficiary can release available tokens
- Admin can revoke unvested tokens (optional if time allows)

4. What we evaluate

- Solidity correctness
- Access control and safety
- Tests quality
- Event design
- README clarity

5. Guidelines

1. This is a 4-hour task
2. Hardhat or Foundry both fine
3. Prefer readable secure code over clever tricks
4. Include how to compile/test

6. Submission

Submit your work from this repository (or your fork), including:

- Branch name used (scratch or boiler-plate)
- Install / compile / test commands
- 5-10 bullet design notes (assumptions, trade-offs, known gaps)
- Time spent

7. Compensation

Paid take-home. Compensation applies whether or not we proceed.